

Experimental Demonstration of Beam-Steered Optical Wireless Localization: A Single-LED Sensing Modality for 6G Integrated Sensing and Communication

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Abstract—Integrated sensing and communication (ISAC) is a defining feature of the ITU IMT-2030 (6G) vision, in which centimeter-level positioning becomes a native network service rather than an add-on. Optical wireless communication (OWC) is a strong candidate for the 6G optical access layer, and beam-steered transmitters—already investigated for high-capacity, mobility-robust 6G optical links—offer a largely untapped opportunity to deliver high-accuracy localization from the very same infrastructure. Building on our model-based beam-steered single light-emitting diode (LED), single-photodiode (PD) positioning framework, so far validated only in simulation, this paper reports its first experimental demonstration on an automated three-dimensional (3D) robotic testbed. A single near-infrared LED is steered through K predefined orientations while a PD held at an arbitrary, fixed orientation records the received-signal-strength (RSS) profile; the transmitter-to-receiver direction is recovered by ratio-based least-squares estimators that are provably independent of the receiver orientation, and the range is obtained either *cooperatively* (a single beam-aligned measurement) or in *broadcast* mode (from the same K measurements). We further introduce an empirical quasi-Lambertian radiation model, calibrated from a dedicated radiometric campaign, and embed it into the estimators to compensate for the angular asymmetries of commercial off-the-shelf LEDs. Over a densified grid of $\approx [2000]$ positions spanning a $[3.0 \times 3.0 \times 1.4]$ m workspace, the system attains a mean direction-finding error of $[X.X]^\circ$ and a 3D positioning error of $[X.X]$ cm (90th percentile $[X.X]$ cm), degrading by less than $[X]\%$ under receiver tilts up to $[30]^\circ$. These results experimentally confirm that beam-steered OWC can meet the IMT-2030 indoor positioning target with minimal receiver hardware, establishing it as a practical sensing pillar for 6G optical ISAC.

Index Terms—6G, integrated sensing and communication (ISAC), optical wireless positioning, beam steering, LiFi, Cramér-Rao bound, receiver-orientation independence, radiation-pattern calibration, experimental demonstration.

I. INTRODUCTION

[I-A. The 6G / ISAC opening — borrow the framing of [1] and [2]]. Open at the top of the wireless-systems hierarchy: while 5G is still being rolled out, standards bodies (ITU) have defined the IMT-2030 (6G) vision, in which integrated sensing and communication (ISAC) is a headline capability [2], [3]. State that the sensing function explicitly includes positioning,

for which IMT-2030 targets an accuracy in the 1–10 cm range, and that localization is now recognized as a *key enabler* of 6G rather than a peripheral service [4], [5]. This paragraph reframes the whole contribution away from “just indoor 3D OWP” and toward “a sensing pillar for 6G.”

[I-B. Why optical, and why the transmitter side]. Motivate the optical spectrum as one of the promising 6G bands, and OWC/LiFi as a mature dual-use medium where illumination, data, and positioning can coexist [6], [7]. Then introduce the central technological trend: 6G optical access is moving toward *transmitter-side spatial diversity and beam management*. Two complementary families illustrate this: (i) dense VCSEL/grid-of-beam arrays with adaptive divergence control for coverage and capacity [8], [9], and (ii) single-beam steering via MEMS micromirrors, liquid-crystal deflectors, or optical phased arrays [10]–[13]. Key rhetorical move: the same beam-steering / beam-management hardware being deployed for 6G optical *communication* can be re-used for *sensing*—this hardware reuse is exactly the ISAC premise. Note the contrast with the VCSEL-array line of work (e.g., [9]): those papers optimize divergence for coverage; we exploit the angular RSS diversity created by steering for localization.

[I-C. The single-LED OWP landscape and our beam-steered paradigm]. Condense the single-LED positioning survey argument [14]–[16]: single-source OWP reduces infrastructure but classically pushes complexity to the receiver (tilted-PD arrays, mechanically rotating PDs, IMU fusion) [17]–[22]. Contrast our paradigm: shift spatial diversity to the transmitter through K steered orientations, keeping the receiver a single static PD. Cite our own prior model-based results—direction finding + cooperative distance recovery [23], receiver-orientation independence and tilt robustness [24], and the broadcast (K-only) distance recovery / multi-user variant [25] (GLOBECOM). Use a near-infrared LED so the positioning link is decoupled from illumination standards [14].

[I-D. The gap this paper fills]. State the gap plainly: all of the above beam-steered OWP results are *simulation-based* and assume an *ideal Lambertian* source and *perfectly known* noise statistics. Two things are missing for 6G credibility: (i) a real hardware demonstration at room scale with ground truth, and (ii) a treatment of the real, quasi-Lambertian / asymmetric radiation pattern of COTS LEDs [26], which biases RSS-only estimators. This motivates an *experimental* paper with an empirically calibrated channel, aligned with related ISAC-oriented calibration work [27], [28].

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[I-E. Contributions]. Present as an itemized list (draft below).

- **First experimental demonstration** of beam-steered single-LED / single-PD 3D optical localization on an automated robotic testbed, with UR5-referenced ground truth over a densified grid of $\approx [2000]$ positions (vs. the ~ 300 -point conference proof-of-concept).
- **Empirical quasi-Lambertian radiation model** for COTS LEDs, calibrated from a dedicated radiometric campaign and embedded into the direction-finding and distance-recovery estimators to remove peripheral (angular) bias.
- **Experimental validation of the two distance-recovery regimes:** cooperative ($K+1$ beam-aligned) vs. broadcast (K -only), quantifying the accuracy/latency trade-off on real data and against the theoretical PEB/PEB_B.
- **Experimental confirmation of receiver-orientation independence** of the direction-finding stage under controlled PD tilts, validating the $\mathbf{n}_r$ -agnostic property predicted in [23], [24].
- **A 6G-ISAC positioning benchmark:** mapping the measured accuracy, robustness, and latency onto the IMT-2030 KPI, and outlining the integration path with an OWC data link.

[I-F. Explicit novelty vs. GLOBECOM and TCOM — REQUIRED paragraph]. Write one crisp paragraph: “Compared with our conference paper [25], which introduced the broadcast concept in simulation only, and with [23], which developed the model-based direction/position estimators and bounds in simulation, this paper contributes (i) the first hardware realization and measurement campaign; (ii) an empirically calibrated (non-ideal) radiation model absent from both; (iii) experimental head-to-head of cooperative vs. broadcast recovery; and (iv) an explicit repositioning of the architecture within the 6G ISAC / IMT-2030 framework.”

[I-G. Paper organization]. Standard closing paragraph referencing Sections II–V.

II. BEAM-STEERED OPTICAL WIRELESS LOCALIZATION FOR 6G ISAC: WORKING PRINCIPLES

A. System Model

Recap the geometry: ceiling-mounted steerable LED at $\mathbf{t} = [0, 0, H]^T$ steered along $\{\mathbf{n}_{t,i}\}_{i=1}^K$; single PD at unknown $\mathbf{r}$ with orientation $\mathbf{n}_r$. State the Lambertian LOS RSS model, the irradiance/incidence cosines $\cos \phi_i = (\mathbf{n}_{t,i} \cdot \mathbf{d})/d$, $\cos \psi = -(\mathbf{n}_r \cdot \mathbf{d})/d$, the compact factorization $\mu_i = \eta Q_i^m$ with $\eta = C \cos \psi / d^2$, and the N -sample MVU mean and Gaussian noise model. Cite [23], [29]. Explicitly flag which quantities will later be replaced by empirical calibration (m , C , and the shape $\cos^m \phi \rightarrow R(\phi)$).

B. Direction Finding (Receiver-Orientation-Agnostic)

Summarize the three estimators from [23]: closed-form GLS and lightweight WLS (ratio-based, 3×3 eigenproblem) and iterative NLS (normalized Lambertian fit on $\mathbb{S}^2$). State the key property to be tested experimentally: all three cancel the common amplitude and are mathematically independent of $\mathbf{n}_r$, [23], [24].

C. Distance Recovery: Cooperative vs. Broadcast

Two regimes tested on the same dataset: (a) *Cooperative / active receiver*: LED steered toward the estimated direction and PD reoriented toward the LED (emulated by the UR5 end-effector) for a dedicated ($K+1$)-th measurement—this is the [23] TCOM pipeline. (b) *Broadcast / passive receiver*: the PD stays static and d is recovered in closed form from the same K measurements via the profiled amplitude estimator $\hat{\eta} = \sum_i \hat{\mu}_i \hat{Q}_i^m / \sum_i \hat{Q}_i^{2m}$ and $\hat{d} = \sqrt{C \cos \psi / \hat{\eta}}$ —this is the [25] GLOBECOM pipeline, enabling simultaneous multi-user positioning. Note that broadcast recovery uses the recorded UR5 pose so a single dataset supports both pipelines with no extra measurement.

D. Empirical Radiation Modeling for Estimation Enhancement

Contrast the ideal $\cos^m(\phi)$ profile with the empirical, quasi-Lambertian / azimuthally-asymmetric pattern $R(\phi, \alpha)$ of COTS LEDs [26]. Propose the integration framework: replace Q_i^m by a calibrated response $\hat{R}(\phi_i)$ (spline look-up table or multi-Gaussian fit) inside GLS/WLS/NLS and inside the $\hat{\eta}$ estimator, and recompute the radiometric constant C under the calibrated profile. Discuss how this removes the spatial (peripheral) bias that a mismatched Lambertian order introduces in RSS-only estimation.

III. EXPERIMENTAL FRAMEWORK AND IMPLEMENTATION

A. 3D Robotic Testbed and Hardware Upgrades

Describe the testbed: steerable NIR LED on a two-axis motorized gimbal (Thorlabs PRM1Z8, 0.1° resolution) at the ceiling; PD carried by a UR5 arm acting as programmable ground truth. List the upgrades over the conference setup: (i) *Thermal management* — aluminum heat sink + star-PCB mount enabling nominal 1 A drive (vs. 750 mA), improving SNR; (ii) *Receiver front-end / FOV* — baseline narrow-FOV PD (BPX 61, $\approx 60^\circ$) vs. wide-FOV upgrade (S6967, $\approx 90^\circ$), and the corresponding reachable workspace (2.4×2.4 m vs. 3.0×3.0 m); (iii) *High-density grid* — 0.2 m resolution yielding up to $\approx [2000]$ positions across heights $h \in [[0.2, 1.4]]$ m. Include a hardware/parameter table (LED power, $\Phi_{1/2}$, A_{det} , FOV, TIA gain, DAQ rate).

B. Kinematics and Coordinate Transformations

Give the rigid-body transforms needed for ground truth: (a) *Transmitter kinematics* — rotation matrices of the two-axis gimbal mapping commanded $(\phi_{\text{cmd}}, \alpha_{\text{cmd}})$ to $\mathbf{n}_{t,i}$; (b) *Receiver kinematics* — mapping the desired PD pose in the global testbed frame to the UR5 base/tool frames, and extracting $\mathbf{n}_r$ from `pose_UR5` (with a note on sign/convention vs. the theoretical model). Document UR5 repeatability (± 0.03 mm) as the ground-truth uncertainty.

C. Radiometric Calibration Protocol

Two calibration sub-datasets from `dataset_specification.md`: *Sub-dataset 1* — $R(\phi)$: at fixed known d , sweep the irradiance angle $\phi \in [0^\circ, 90^\circ]$ (step 2–5°) over several azimuth cuts (0/90/180/270°) to capture asymmetry; record raw $V_{\text{raw}}[n]$, dark $V_{\text{dark}}[n]$, driving current I_{LED} , temperatures. Fit $\hat{R}(\phi)$ (spline / multi-Gaussian). *Sub-dataset 2* — constant C : PD under the LED (nadir/zenith) at known distances (0.4–1.2 m); report C under the calibrated $\hat{R}$ vs. the ideal-Lambertian C .

D. Spatial Data-Acquisition Protocol (Main Campaign)

Describe Sub-dataset 3: for each of the $\approx [2000]$ points, record ground truth (x, y, z , full `pose_UR5`, $\mathbf{n}_r$); for each of the $K=9$ TCOM-optimized orientations record $\mathbf{n}_{t,i}$, raw/dark samples ($N=1000$ @ 1 kHz) and timestamps; plus the cooperative ($K+1$)-th beam-aligned measurement. Repeat the full $\{K, K+1\}$ scan $M \in [10, 30]$ times per point for empirical noise statistics ($\hat{\sigma}^2$), enabling direct comparison with the σ^2 assumed in [23], [25]. Emphasize that storing `pose_UR5` lets the *same* data feed both cooperative and broadcast pipelines.

E. Receiver-Tilt Robustness Protocol

Describe the deliberate-tilt sweep (axis D2): on a subset of $[100-200]$ points, repeat the $\{K\}$ scan with the PD tilted to $\theta_{\text{tilt}} \in \{0, 6, 15, 30\}^\circ$ over varied tilt azimuths, replicating the PEB-vs-tilt analysis of the broadcast paper. This isolates the experimental impact of physical attitude on both direction finding ($\mathbf{n}_r$ -agnostic) and distance recovery ($\mathbf{n}_r$ -dependent).

F. Dataset Structure and Reproducibility

Specify the storage schema (one row per (`point_id`, `orientation_id`, `repeat_id`), separate tables for ($K+1$) and for the calibration sub-datasets), retention of raw $V_{\text{raw}}[n]$ for $\hat{\sigma}^2$ recomputation, and session metadata for traceability. State the intention to release the dataset (open-data / reproducibility angle, valued for JLT and for the 6G community).

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Empirical Radiation Pattern Characterization

Present the measured $\hat{R}(\phi)$ (and azimuthal asymmetry) vs. the best-fit ideal $\cos^m \phi$; report the fitted effective m , the RMS model mismatch, and the corrected vs. ideal C . This is the empirical foundation used by the later capstone subsection.

B. Baseline Direction Finding and 3D Positioning

Report GLS/WLS/NLS direction error (mean, median, P_{90}) and full 3D error (RMSE, P_{90} , APE) over the dense grid, for cooperative and broadcast recovery, compared against the theoretical DEB / PEB / PEB_B from [23], [25]. Include spatial heatmaps and CDFs. Discuss where measured error departs from the bound (peripheral bias $\rightarrow$ motivates the next subsections).

C. Broadcast vs. Cooperative Distance Recovery

Head-to-head on identical data: accuracy gap and, crucially, the operational trade-off (broadcast = passive, simultaneous multi-user, no PD reorientation; cooperative = one extra aligned measurement, higher accuracy). Report empirical latency/robustness. Tie back to the multi-user 6G use case (multi-robot / AGV / IoT).

D. Robustness to Receiver Tilt ($\mathbf{n}_r$ -Agnosticism)

Use the tilt-sweep data to show experimentally that direction finding is (near-)invariant to θ_{tilt} while broadcast distance recovery degrades in the predicted way; quantify degradation vs. θ_{tilt} and vs. K . This validates the theoretical $\mathbf{n}_r$ -independence and the tilt sensitivity of the amplitude term.

E. Performance Enhancement via Empirical Radiation Modeling

Compare baseline (ideal-Lambertian) estimation against the estimation that plugs in the calibrated $\hat{R}(\phi)$. Show the reduction in peripheral direction error and 3D RMSE; report the percentage improvement and the fraction of positions moved below the 1–10 cm IMT-2030 band. This is the headline experimental contribution.

F. Discussion: Implications for 6G Optical ISAC

Map results onto IMT-2030 KPIs (accuracy 1–10 cm, latency), argue the hardware-reuse ISAC synergy (same steered transmitter serves data + localization), and honestly scope the claim: this paper demonstrates the *sensing/localization pillar*; joint data transmission is the natural next integration step (not demonstrated here). Contrast positioning-oriented steering with coverage-oriented VCSEL divergence control [8], [9].

V. CONCLUSION AND FUTURE WORK

Summarize: first experimental demonstration of beam-steered single-LED / single-PD 3D optical localization, empirically-calibrated radiation modeling, cooperative-vs-broadcast validation, and tilt robustness—together meeting the IMT-2030 centimeter-level target with minimal receiver hardware, positioned as the sensing pillar of 6G optical ISAC. Future work: (i) true joint communication + localization on the same steered link (full ISAC, e.g., m-CAP data on the NIR carrier [1]); (ii) non-mechanical / high-speed steering (liquid-crystal, OPA) for mobility [10], [11]; (iii) IMU-orientation-error mismatch analysis for broadcast mode; (iv) scaling to multi-AP and VCSEL-array transmitters [8], [9].

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